

Rudra Garg

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EDUCATION

B. Tech in CSE	Indian Institute of Information Technology Guwahati	2022–2026	8.42 (Sem 7)
Class XII	C.B.S.E. K.D.B. Public School, Ghaziabad	2022	93.6%
Class X	C.B.S.E. K.D.B. Public School, Ghaziabad	2020	95.2%

SKILLS

- **Languages:** Python, Go, JavaScript, C++, Java
- **Technologies:** Flask, FastAPI, Vue 3, React, Gin, Flutter, Docker, Kubernetes, Temporal, Prometheus, Grafana
- **Cloud/Databases:** Google Cloud, Firebase, AWS, Azure, Supabase, PostgreSQL, MySQL, MongoDB
- **Tools:** Git, GitHub, Figma, Postman, Swagger UI, Confluence, Jira, Linux

WORK EXPERIENCE

Back End Developer Intern – GRIG Technologies (June 2024 – August 2025)

- Led design and implementation of a certificate generation/management system (**Flask, Firebase**), reducing processing time by **98.3%** and increasing data retrieval efficiency by **40%**.
- Directed the evaluation of **Supabase** for self-hosted object storage, prototyping solution with **Flask, SQLAlchemy**, and **Flask-RESTx** to assess performance, scalability, and enterprise fit.
- Architected and developed **10+ RESTful APIs (Flask-RESTx)** for user/product management, incorporating custom validators and exception handling to ensure data integrity.
- Engineered scalable alert microservice (**FastAPI, SQLAlchemy**) for Ventory, integrating **Zoho OAuth2** and **Temporal.io** for orchestrating complex, stateful email notifications.
- Spearheaded containerization (**Docker**) and streamlined deployment (**Kubernetes**), defining **CI/CD** workflows (**GitHub Actions, K8s manifests**) to enhance scalability and efficiency.

PROJECTS

Portal Gambit – Chess Variant Game | [Github-Link](#)

- **Collaborated and led a team of 4** in the end-to-end development of full-stack, real-time chess variant (**FastAPI, React/Vite**), integrating **Firebase Realtime Database** for synchronization and user authentication.
- Devised a custom PortalChess engine on the frontend, implementing unique portal mechanics, complex move validation, and **WebRTC** voice chat (**PeerJS**) for enhanced player interaction.
- Designed and built robust backend services (**FastAPI, Firestore**) for game analytics (**10+ metrics**), user management, and history, secured with **JWT** authentication.
- Automated deployment to **Google Cloud Run (Docker, Cloud Build CI/CD)** reducing manual intervention and achieved comprehensive test coverage with **15+ Pytest suites**.

TerraQuest – Geography Guessing Game | [Github-Link](#)

- Created an interactive geography guessing game using **Vue 3/Vite, Pinia**, integrating **Google Street View** and **Leaflet.js** for interactive map guessing.
- Developed a robust **Go** backend API (**Gin Gonic**) with **JWT** auth (**bcrypt**), game session management, and **PostgreSQL** persistence (**GORM**), documented via **Swagger/ReDoc**.
- Implemented real-time multiplayer (up to **8 players**) using **Go/WebSockets (Gorilla WebSocket)**, enabling synchronized lobbies, status, chat, and round-based gameplay.
- Configured automated frontend deployment (**GitHub Actions CI/CD**), containerized backend (**Docker**), and built concurrent **Go** script to populate DB by validating **81,000+ locations** via **Google Street View API**.
- Managed comprehensive frontend state using **Pinia** for auth, single-player flow, and complex real-time multiplayer interactions.

Fog Computing Module Placement Research with Docker Testbed | [GitHub-Link](#)

- Constructed a **4-tier** distributed Fog computing testbed (**Python/Flask** microservices) using **Docker/Docker Compose (15+ containers)** to evaluate module placement strategies for EEG processing.
- Simulated realistic network emulation (latency up to **300ms**, jitter **30ms**, packet loss) using **Linux Traffic Control** within Docker, enabling accurate performance testing.
- Integrated monitoring stack (**Prometheus, Grafana, Loki/Promtail**) for metrics (CPU, latency, throughput) and logs, providing observability into distributed system behavior.
- Evaluated placement strategies (GVMP vs. EWMP), demonstrating significant performance impact with up to **95% network reduction** and quantifiable latency differences (e.g., Mobile **~16ms** vs. Cloud **~5600ms p95**).

Pilgrim AI – Mental Health Support Platform | [GitHub-Link](#)

- Formulated an **AI-driven chatbot** for **24/7 confidential mental health support**, enhancing accessibility and reducing stigma.
- Crafted a cross-platform mobile app using **Flutter/GetX** for seamless, responsive user experience.
- Deployed the **Flask** backend on **Google Cloud App Engine** to handle secure, reliable API requests and user Interactions and a **99.9% uptime**.
- **Enhanced AI models (Vertex AI, LlamaChatBison2)** to facilitate natural conversations, enhancing user engagement and support quality.

PROFESSIONAL CERTIFICATIONS

- Google Cybersecurity Professional Certificate – Coursera
- Google Cloud Computing Foundations & Generative AI – Google Cloud Study Jams 2023, NPTEL
- Cybersecurity, Web Development, and AI Fundamentals – IBM Skills Build
- Programming and Data Structures (Python, C++) – NPTEL, Coding Blocks, HackerRank